MILITARY SPECIFICATION SHEET

FIXTURES, LIGHTING; FLUORESCENT, EXTENSION LIGHT FOR SERVICING ELECTRONIC EQUIPMENT, 4 WATTS, 120 VOLTS, 60 HERTZ, SYMBOL 163

This specification sheet is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the fixtures described herein shall consist of this document and the latest issue of MIL-F-16377(SH).

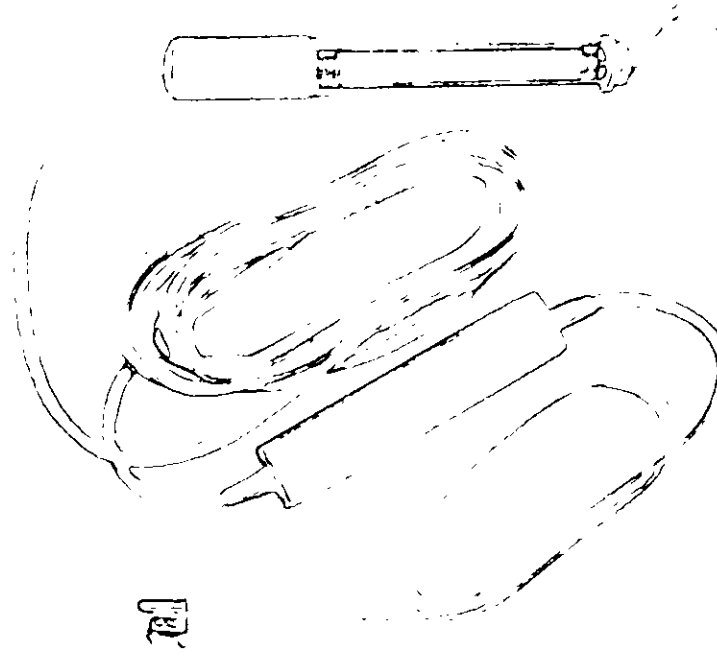


FIGURE 1. Typical view of Symbol 163.

All dimensions in inches.
For tolerances, other than specified herein, see paragraph 3.5.2 of MIL-F-16377.

PSC 6230

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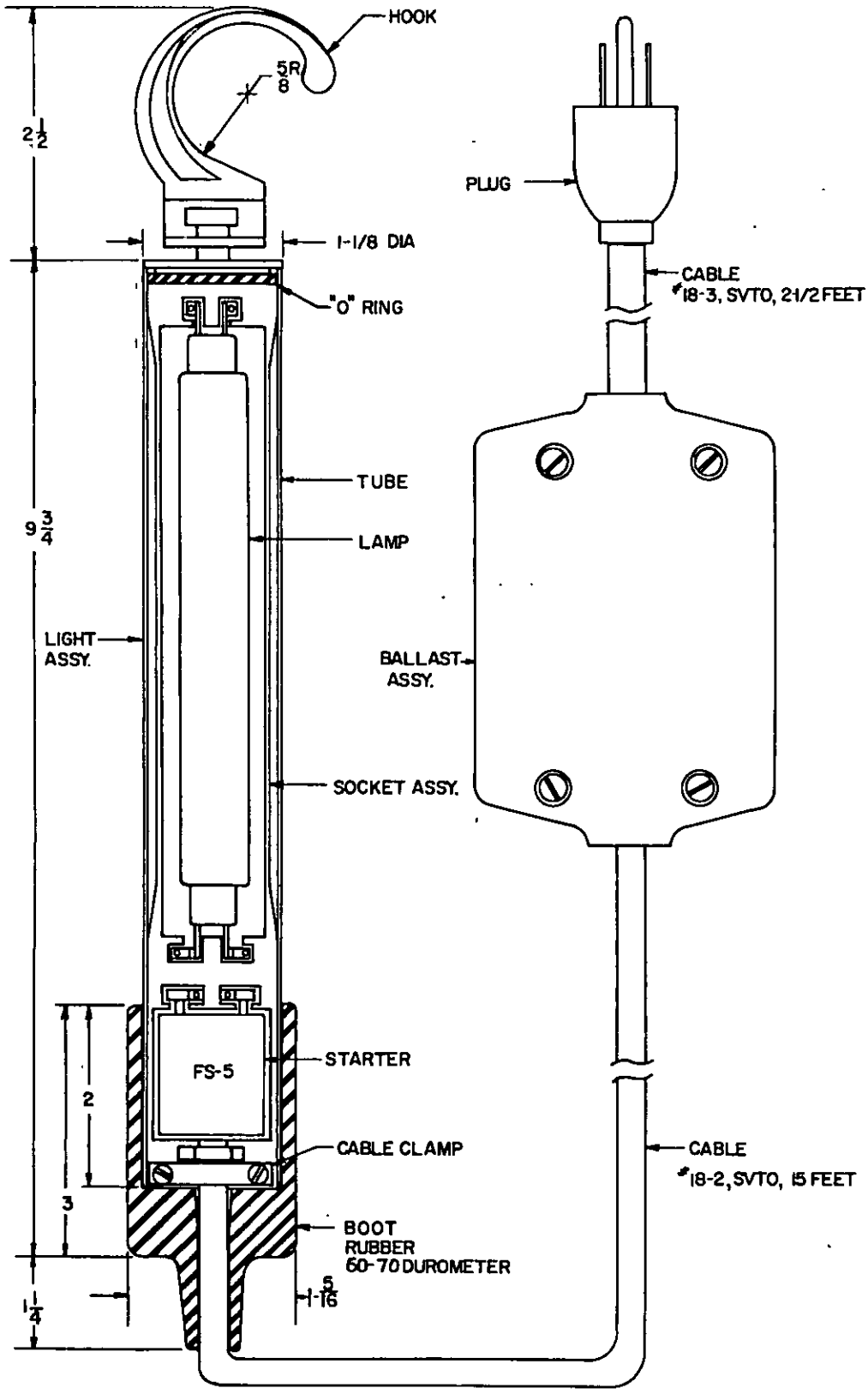


FIGURE 2. Extension light assembly.

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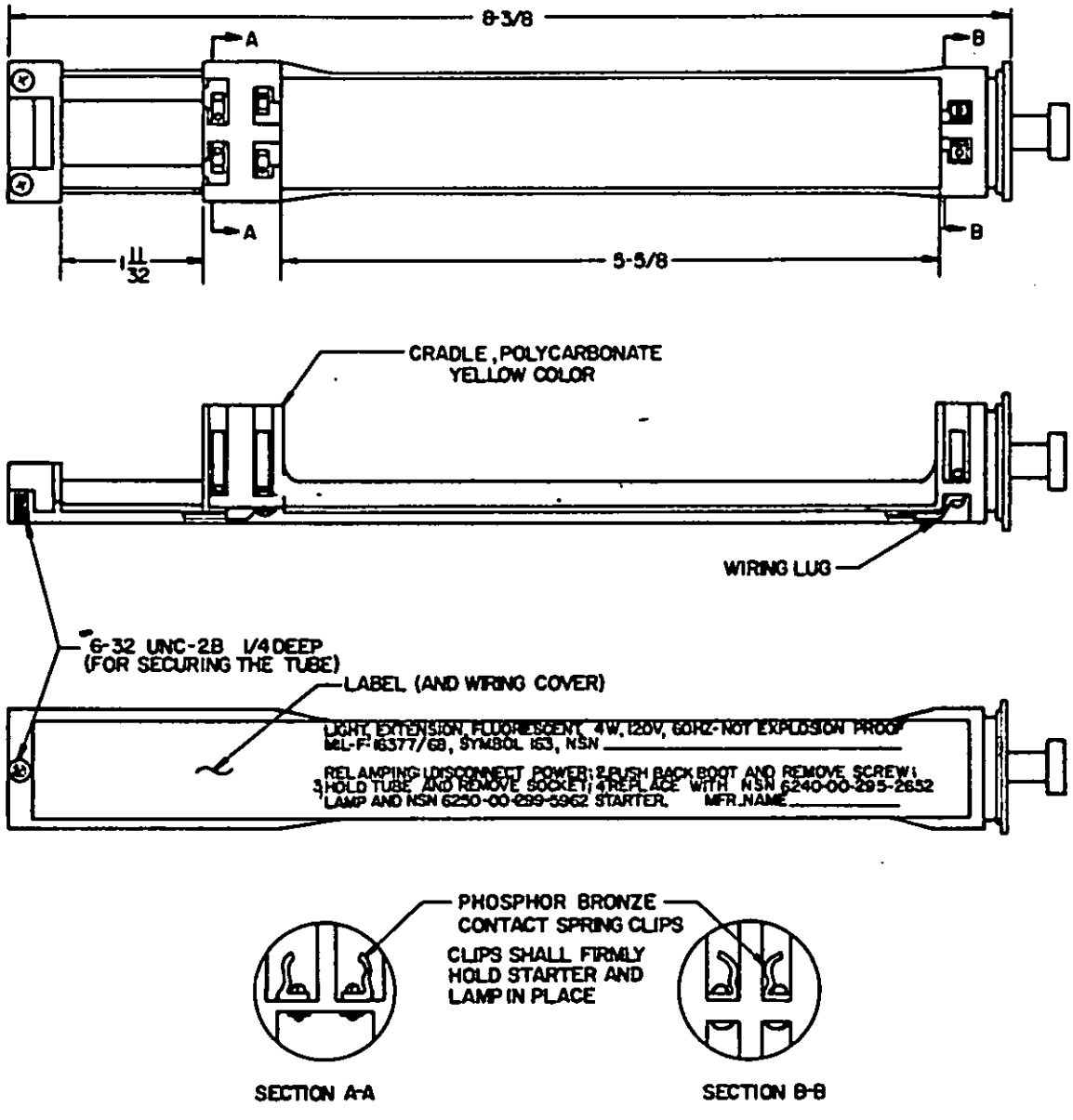


FIGURE 3. Socket assembly.

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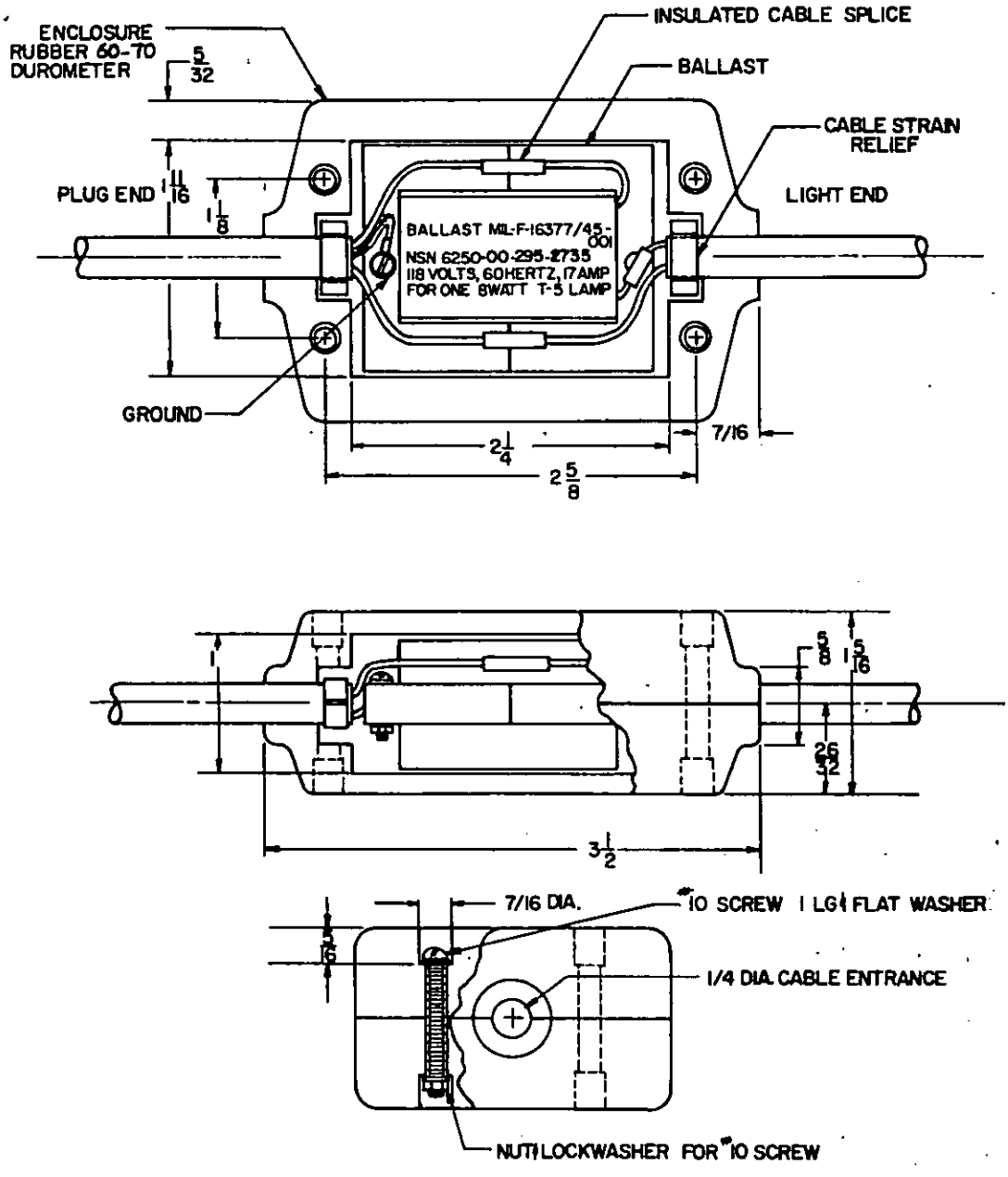


FIGURE 4. Ballast assembly.

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MATERIAL SHALL BE CLEAR POLYCARBONATE, UV STABILIZED, WITH A MINIMUM 90% LIGHT TRANSMITTANCE.

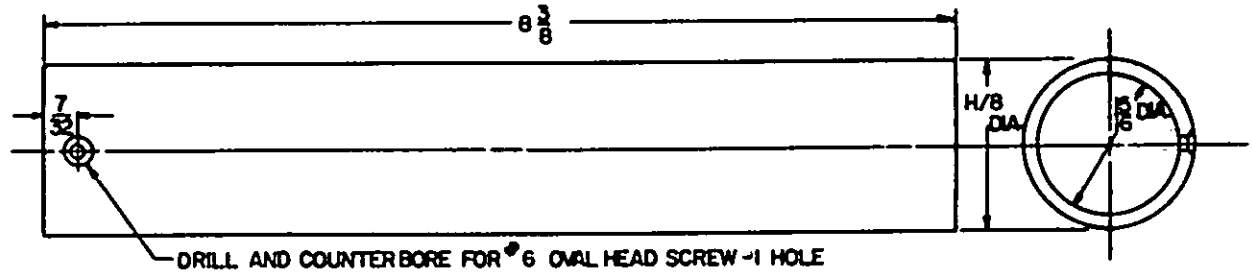


FIGURE 5. Tube.

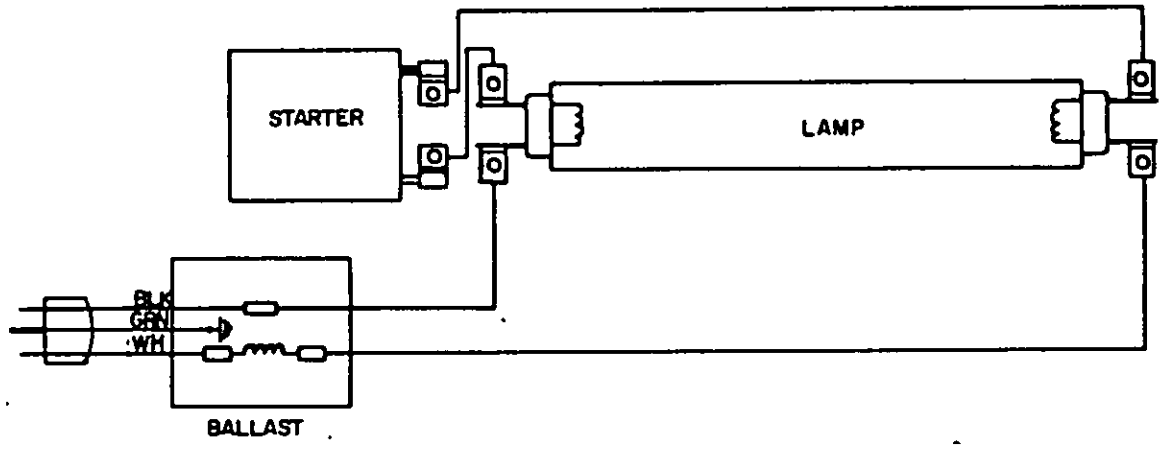


FIGURE 6. Wiring diagram.

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REQUIREMENTS:

Type: I
Class: 1
Symbol 163: Light, extension, fluorescent, 4 watt, 115 volt, 60 Hertz.
Design:
Electrical rating: 115 volts, 4-watt, 60 Hertz
Dimensions and configurations: See figures 1 through 6.
Light assembly: The light shall be designed to be dripproof and shall be so constructed that no parts will work loose in service. The light assembly shall have no exposed metal parts except for the ballast mounting screws which shall be recessed.
Lamp: Lamp shall be fluorescent, 4 watt, 115 volt, Industry Number F4T5/CW, NSN 6240-00-295-2652.
Starter: Starter shall be type III of W-S-755, Industry Number FS-5.
Ballast: Ballast shall be part number M16377/45-001 of MIL-F-16377/45.
Cable: Cable size, type, and length shall be as shown in figure 2.
Plug: Plug shall be 2-pole, bladed, 3-wire grounding (NEMA number 5-15P) and it shall be attached at the end of the cable as shown in figure 2. The grounding contact shall be electrically connected to ballast frame.
Hook: Material for hook shall be polypropelene. Attachment to the light assembly shall be such that the hook can be removed by the user if so desired.
Cable pull: Strain relief shall be provided so that a mechanical stress on the flexible cable will not be transmitted directly to binding-screw terminals, splices, or internal wiring. The assembly of the cable to a fitting shall be capable of withstanding without displacement, a straight pull of 30 pounds applied between the fitting and the cable for a period of one minute.
Enclosure effectiveness: The light shall be designed to be dripproof, 15 degrees.
Drop: The light shall not be impaired when dropped to a concrete floor ten times from a height of 3 feet. The light shall be allowed to freely fall in a manner that would normally be the case if the light were to fall from a horizontal platform to the floor.
Weight: 2.5 pounds maximum.
Identification: The information on the label shall be as shown on figure 3.
Quality Assurance: Quality assurance shall be as specified in MIL-F-16377 and as follows:
First article inspection: See table I.
Quality conformance:
Comparison inspection: Same as first article inspection.
Inspection of product for delivery:
Examination: See MIL-F-16377.
Examination for preparation of delivery: See MIL-F-16377.
Quality conformance tests: See MIL-F-16377.

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TABLE I. First article and comparison inspection for complete lights.

Inspection	Requirement paragraph	Inspection paragraph	Remarks
Examination	3.13	4.5	
Operation	3.13	4.8.1	
Dielectric withstanding voltage	3.6.15	4.8.2	
Insulation resistance	3.6.16	4.8.3	
Enclosure effectiveness	3.5.11	4.8.14.2	
Dielectric withstanding voltage	3.6.15	4.8.2	
Insulation resistance	3.6.16	4.8.3	
Cable pull	1/	1/	
Grounding circuit	3.6.13.3	4.8.11	
Leakage current	3.6.13.5	4.8.13	
Optical uniformity	3.7.4.1.1	4.8.4	
Stress relief	3.8.5.3	4.8.5	

1/ As specified herein.

General Information: (a) This light is intended to be used as a service light inside confined spaces of electronic cabinets onboard ship.

(b) Assignment of Military part numbers:
Complete light: M16377/68-163
Tube: M16377/68-001 .

Preparing activity:
Navy - SH
(Project 6230-N271)

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